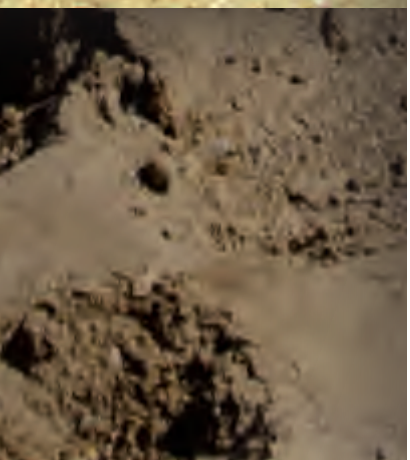
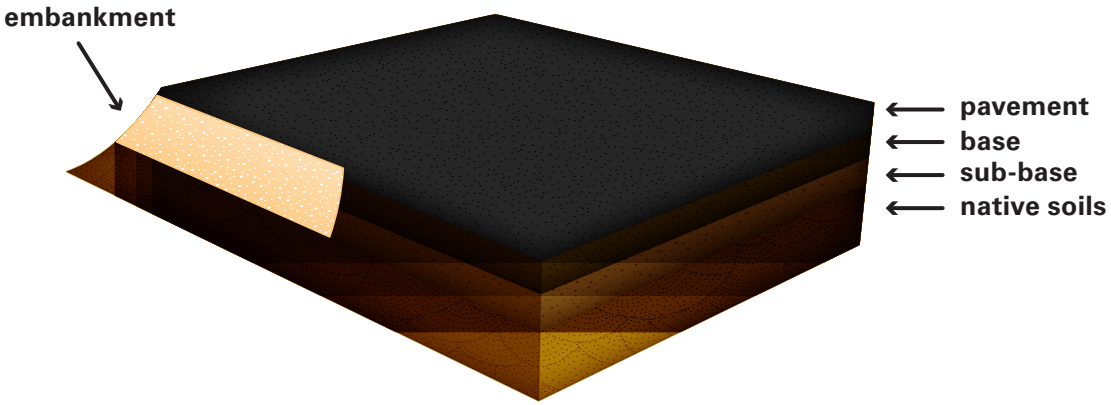






Unit 4
APPLICATION AND CONTROL OF QUALITY

Project parameters have a significant impact on the placement of soils and their compaction. Learning to maximize results under wide ranges of conditions will help you maximize efficiencies and avoid rework. Time-tested and new methods of measurement are also beneficial to helping you better manage your compaction projects.



[THE ELEMENTS OF JOBSITE STRUCTURE]

Like any kind of structure, a road is composed of specific components that perform specific functions. Terminology can vary regionally, but these components work together to support the load of the traffic upon it.

Natural or Native Soils – Sometimes called the "substrate," "subgrade" or "basement soil," this is the foundation of a roadbed and consists of the soils and materials that are naturally occurring without human or chemical modification. During the process of building a road, the surface soils are stripped away to provide a level, even grade. The exposed materials at the bottom of the cut are the natural soils. If these existing materials are insufficient to support the load of the road structure, they are modified or replaced with suitable materials. Modification can be mechanical, which includes compaction, reinforcement with geosynthentics or the incorporation of aggregates; or chemical, which includes the incorporation of binders such as Portland cement; or a combination of mechanical and chemical. The goal is to improve the load-bearing capabilities of the material. Ultimately, this foundation must provide adequate support for the structure upon it.

Embankment – An embankment is any fill where the top is higher than the adjoining surface. The roadbed is designed to be a certain width and support the road at an engineered elevation and grade. In some cases, naturally occurring undulations of the terrain may require the use of fill material to provide an adequate subgrade for the roadbed. Embankments are constructed for this purpose by placing and compacting appropriate aggregate materials to build up the terrain until it meets the required elevation.

Sub-base – The primary functions of this layer are to distribute the load of the structure it supports into the substrate and provide a relatively even grade upon which to place the base course. However, the sub-base also can provide a number of additional functions depending on the composition of the substrate materials, such as filtering functions or acting as a barrier against capillary water. Typically, this layer is composed mostly of substrate materials that are modified if necessary and compacted. There can be multiple sub-base layers, as secondary lifts can be added if required to support greater loads. Generally, the aggregates used in the sub-base will be larger than the aggregates used in layers the sub-base supports.

Base – The base layers serve a similar function to the sub-base. They also distribute the load they carry and provide protection from the effects of water and frost. There can be a single base or multiple base layers depending on the load requirements. The base is composed of an engineered gravel mix with a particle size smaller than that used in the sub-base.

Asphalt layers are placed upon this foundation. The quality of the foundation will affect the durability of the asphalt layers placed upon it. For more information about the asphalt layers, consult Caterpillar's companion "Guide to Asphalt Compaction."

[SOIL COMPACTION PROCEDURES]

JOB LAYOUTS

Before laying out a compaction job, the main questions a contractor must consider are:

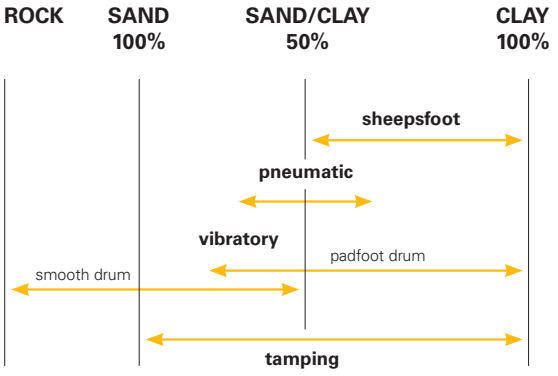
- What is the soil gradation and classification?
- What is the maximum dry density and optimal moisture content?
- What is the compaction requirement (%)?
- What are the compactor settings and speed?
- What is the lift thickness?

A contractor needs to know the material, the requirements and the application for each type of equipment. Once these variables are understood, the contractor can begin to think about the best approach to the job, and may also consider available technologies to apply.

Compactor Application and Sizing – Operating characteristics for the various types of compactors are shown in the accompanying illustration. For comparison, the illustration below shows application ranges where each compactor is most effective. The machines may overlap in these ranges and it is not uncommon to see machines working on materials outside their normal application zone. Therefore, the information presented in these charts should be considered strictly as average application guidelines.

Sitework and Embankment Applications – Sitework refers to a broad range of earthen construction, including: foundation preparation, grading, backfilling and basin filling. It could be a building site, a road, or some other surface or structure.

APPLICATION CHART



OPERATING CHARACTERISTICS FOR SOIL COMPACTION EQUIPMENT

Self Propelled Only

Machine	Sheepsfoot	Pneumatic (15 ton and up)	Tamp Foot	Vibratory
compacted lift thickness - mm (in)	150 - 300 (6 - 12)	150 - 300 (6 - 12)	150 - 300 (6 - 12)	150 - 600 (6 - 24)*
average working speed - km/hr (mph)	6 - 10 (4 - 6)	6 - 19 (4 - 12)	16 - 32 (10 - 20)	2 - 8 (2 - 5)
cycles (1 cycle = 2 machine passes)	6 - 10	3 - 8	4 - 8	2 - 4

**Depends on compactor size and compaction target*

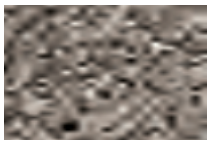


Rock Fill – Rock is increasingly used as embankment fill in highway construction. It is also used to a greater extent in dam, airport, building and harbor embankment construction. Shot rock often contains so many fines that considerable settling will occur if the fill is not compacted.

Rock fill is usually spread in 450-1200 mm (18-48 in) lifts. How the material is spread before compaction is vital. Tractor spreading in layers creates uniform fill because the dozer blade does some reorienting of the rocks and the tracks provide some compaction. Therefore, a relatively dense and even surface is prepared for the compactor.

Heavy compaction forces are needed after spreading to relocate large stones for density and stability. The largest smooth drum vibratory compactors are selected for this job. Even so, compactors are subjected to great stresses on rock fill. The drum should be constructed of thick, high-grade steel. If there is a crushing effect on the surface material, the number of passes may have to be reduced. Or, if the machine is equipped with more than one amplitude setting, lower amplitude can be used to reduce surface material distortion.





Sand and Gravel – Vibratory compaction with smooth drum machines is especially suitable and economical on sand and gravel. High densities can be achieved in few passes with the lift thickness determined by the size of the compactor.

Free-draining sand and gravel that contain less than 10 percent fines is easily compacted, especially when they are nearly saturated. When high density is required and the lifts are thick, water should be added. This water will drain out of the lift during the compaction process.

If the sand and gravel contain more than 10 percent fines, the soil is no longer free draining and may become elastic when the water content is high. This type of soil has optimum moisture content where maximum density can be reached. Drying of the wet soil may be necessary to reach the optimum moisture content.

On poorly graded sand and gravel, it is difficult to achieve high density close to the surface of the fill. There is low shear strength in poorly graded soils and the top layer tends to rise up behind the drum. This is not a problem when multiple lifts are being compacted. The previous top layer will be compacted when the next layer is rolled. However, the difficulty of compacting the surface should be kept in mind when testing for density.



When moisture content is below the optimum, water may be added to ensure that proper compaction can be reached.



Silt – Silts are non-plastic fines that are usually compacted with smooth drum vibratory compactors. They can be spread and rolled in thick lifts.

Like all fine-grained soils, their compactability is dependent on moisture. For best compaction results, the water content should not vary much from the optimum moisture content. If too much water is present, silts rapidly approach the fluid state and compaction is impossible. This means that the lifts may have to be aerated with discs, mixed with drier soil (an expensive procedure) or the borrow material may need to be drained better. Silty soils that also contain clay may have considerable cohesion. On these soils, padded drum, tamping foot or pneumatic compactors will achieve better results.



Clay – Clays have plastic properties, which means that the compaction characteristics are highly dependent on moisture content. When the water content is low, clay becomes hard and firm. Above the optimum moisture content, clay becomes more and more plastic and difficult to compact.

Often in clay compaction, the main problem is the need to adjust the water content. The addition of water by using water trucks, discs or soil stabilizers is time-consuming. Water infiltration into the borrow pit may be a better alternative. Drying wet clay can only be done in warm and dry conditions, even using discs and soil stabilizers. Prolonged rolling with sheepsfoot compactors is sometimes done to lower the moisture content.

Even at the optimum moisture content, clay requires a higher compactive effort and a thinner lift thickness compared to non-cohesive soils. Padded drum compactors work best because as the pads penetrate the soil, they break the natural cohesive bonds between the particles. Pneumatic compactors can be used on clays with a low to medium Plasticity Index.

On projects where high production is a requirement and clay is used as fill, good results can be obtained by using tamping foot compactors in conjunction with vibratory padded drum compactors. Tamping foot compactors equipped with dozer blades are efficient at spreading the fill and breaking large, hard lumps of clay often found in clay borrow material. These machines perform the first passes. Final density is reached by using vibratory padded drum compactors.



When moisture content is too high, opening up the soil speeds drying time.

Base and Sub-base Applications – Bases and sub-bases are the layers constructed on top of an embankment or natural ground surface. These layers serve as the foundation for placement of a surface structure such as a road or a building. They increase in strength as they near the finish surface. The materials used in these layers depend on the type of loads the road or building must support.

Usually, very strict specifications are given for base and sub-base materials, for the thickness of the lift, the required density, and the modulus of deformation.

Natural (Native) Soils – From an economic standpoint, it is preferable to use locally available soils. If these soils are suitable, they may be used without chemical treatment or additives. Proper compaction of these soils will substantially increase their load-bearing capacities and control other factors such as permeability, capillary action, and shrink and swell.

The choice of compaction equipment will depend on the type of soils. Generally, granular, non-cohesive soils are specified as base and sub-base material. Smooth drum or pneumatic compactors are more often used in this application.

Treated Soils (Soil Stabilization) – Mixing chemicals with natural or imported soils can substantially improve the soil's stability and load-bearing characteristics. This is called soil stabilization.

After lime or cement has been mixed into the soil, the soil should be compacted with a vibratory compactor. The type of compactor used will depend on the soil's original, untreated characteristics; but generally, a vibratory compactor of 15 tons (33,000 lb) or greater should be used. Where a large volume of cohesive soil is involved, a tamping foot compactor may be more economical than a vibratory compactor. Smaller volumes may be compacted with a pneumatic compactor.

At the beginning of the work, it is recommended that you conduct a stabilization suitability test to ensure that the results are sufficient.



Soil Stabilization improves the engineering qualities of a soil.

Crushed Rock – Job specifications may call for well-graded crushed rock to be used as base and sub-base materials. By using crushed material, gradation can be controlled during the crushing process to match specifications. Crushed rock is generally easier to spread and compact than fine soils. Plus, the compaction results are more predictable. However, these compaction advantages are offset by the expense of crushing and the often-longer hauls to the project site.

Crushed rock is usually hauled to the job in end-dump trailers and placed on the grade with a motor grader or spreading machine. The base material is then spread and shaped in lifts ranging from 150-250 mm (6-10 in). After spreading, compaction is accomplished by smooth drum compactors (static or vibratory) or pneumatic compactors.

[SOIL COMPACTION TIPS]

Despite its simple appearance, soil compaction can be one of the most difficult elements of any construction project. Many manufacturers offer options on their compaction equipment that allow the operator to adjust machine vibration, frequency and amplitude to meet job specifications.

There is no easy way of adjusting the working parameters of a compactor according to the material it is compacting. Trial-and-error is often he best method. Obviously, the user will need to select a compactor of the appropriate size (drum width, weight, etc.) to match the production requirements. But, achieving the maximum compactive effort is usually accomplished by experimentation with the variables that the operator can control—frequency, amplitude and rolling speed—then analyzing performance and making adjustments.

Manufacturers are building more technology into their machines that can assist the operator in maximizing efficiency. But, even with increasing technology and sophistication, to achieve the best soil compaction results you may have to review some basic principles of soil compaction that have proven themselves on construction projects for years. Following are some soil compaction tips to provide basic guidance.

WHICH ROLLER FOR THE APPLICATION?

COHESIVE MATERIAL
Thin layers
Single-drum roller (padfoot)



FINE MATERIAL
Water sensitive
Single-drum roller, smooth or padfoot



FRICTION MATERIAL
(sand 0,063-2 mm/0.002-0.07 in)
Free draining if fine content < 7%
Single-drum roller, tandem roller, pneumatic tire roller



COARSE GRAINED, FREE DRAINING
Friction material
Single-drum roller, tandem roller



COARSE GRAINED
(gravel 2-63 mm/0.07-2.5 in)
Free draining
Friction material
Single-drum roller, tandem roller



COARSE MATERIALS
Heavy particles
Large plates, large single-drum roller (>12.7 tons)



	Permeability	Foundation support	Pavement subgrade	Expansive	Compaction difficulty
Gravel	Very high	Excellent	Excellent	No	Very easy
Sand	Medium	Good	Good	No	Easy
Silt	Medium low	Poor	Poor	Some	Some
Clay	None+	Moderate	Poor	Difficult	Very difficult
Organic	Low	Very poor	Not acceptable	Some	Very difficult

Uniformity of Lift, Speed and Rolling Pattern – The structure of buildings and roads depends on soil bases that are not only compacted to specification, but are compacted uniformly as well. Variations in base density can lead to potholes or wheel ruts in roads and foundation settling under buildings. One of the biggest causes of soil density variations is the use of different soil types that are placed side by side.



Because different soil types will compact differently and yield different compaction results, effort should be made during construction to utilize similar material for each lift. If different types of material are required, try to use the same type for each lift—do not use different material in a horizontal spread. This can be critical in achieving uniform compaction results.

Another cause of density variation is inconsistent moisture content. Because it is more difficult to compact soil that is too dry or too wet, the optimum moisture content as determined by the Proctor Test should always be targeted during compaction throughout the entire jobsite. This will help achieve the most uniform and dense results possible.

Lift thickness is usually not tightly controlled, except perhaps for a base course beneath a pavement surface. If all other factors, such as moisture content and material type, are kept constant, uniform lift thickness will lead to uniform density across the jobsite. Neglecting lift thickness could result in failing to meet the job specifications.

Other factors that can affect compaction and are often given inadequate attention are the coverage and pass counts. Parameters such as the number of passes, compactor speed and vibratory settings can be easily controlled. Intelligent compaction technology, such as Cat Compaction Control with GNSS mapping capability, can provide a visual reference to ensure that proper coverage and pass counts are maintained. Uniform coverage is more efficient and will be more successful than randomly compacting the material.

Lift Thickness – Compactor technology is continually advancing and offering more options and variations to the compaction process. However, after equal compaction effort is applied—even with the most advanced compactor—relatively thick lifts of material will be less dense than thinner lifts. There are exceptions to this rule, but a lift thickness should be sought which optimizes production rates based on required density and minimum compactor passes.



The condition of the previous lift or underlying base will also have an effect on achieving compaction. If a sub-layer has not been compacted thoroughly, it will contain areas that are relatively soft. In turn, compaction on the next lift will have varied, undesirable results. Each lift should be uniformly and thoroughly compacted to ensure subsequent lifts can be successfully compacted as well.

Compactive Effort – Compactive effort is the amount of energy imparted to the soil in order to rearrange and compact the soil particles. Varying machine parameters such as weight, width, tire pressure, vibration amplitude and frequency can change compactive effort. Some of these parameters can be adjusted on a single machine. Others, such as width, may require a different machine in order to alter compactive effort. Certain magnitudes of these parameters are necessary for any given project.



Working Speed – In general, travel speed plays an important part in construction productivity; the faster the machine, the faster the job is complete. However, with vibratory compactors, unlike all other types, productivity will generally increase by decreasing the travel speed. There is an optimum economical speed that will allow a compactor to achieve required compaction.



General Rules of Soil Placement and Compaction – Following are general rules that you, your jobsite management and operators should consider at the start of each compaction project.

- When placing a new lift of soil material, spread material over the entire area uniformly. Avoid placing soils that have excessively high moisture content. Spread the material with a slow-moving dozer and shape the appropriate profile. Do not bury saturated layers with new materials.
- Surface cavities or areas with visible segregation should be remedied by adding well-graded material of the same composition.
- Immediately after spreading is complete, compact the soil, starting from outside and working toward the middle of the area.
- Embankment/shoulder areas must be compacted as well; compact the shoulder from the outside edges and work toward the middle areas. Smooth and seal the surface.
- When working with materials that are susceptible to weather, all placed soil should have a side slope of about 6 percent to prevent the accumulation of surface water.
- When poor weather occurs, compact each lift by working a single drum-width strip until it is fully compact. Then move over to the next strip, repeating the process and working your way across the lift until the entire area has been compacted. Be sure to smooth and seal the surface at the end of the day to prevent further water penetration.
- When compacting a flexible base, it is advantageous to use low amplitude with high frequency if the base is adequately rigid. Use high amplitude and medium-to-low frequency when the base is being compacted together with the first lift.
- Using a combination of static and vibratory compaction—vibratory first followed by static—may yield the best results.

[COMPACTION MEASUREMENT METHODS]

Density has been the traditional method of quantifying soil compaction in the lab, and density has also been the most common historical method to set and measure compaction requirements in the field. Laboratory tests (such as Proctor) determine the moisture content at which maximum density can be attained. Field target densities are specified as a certain percent of the maximum laboratory dry density.

Generally, required field densities will be 95 percent of Standard Proctor for embankments and up to 100 percent of Modified Proctor for roadway structures. Likewise, the moisture content must be within a range of the laboratory determined optimum moisture content.

Point-by-point field-testing has long been required to ensure that the two important elements—target density and moisture content—are being maintained to specification throughout a particular construction job. These tests can also indicate the effectiveness of the compaction equipment and construction methods being used. Recently, machine-integrated compaction measurement methods have become more accepted as a means to determine the quality of compaction on the jobsite.

Many authorities are now requiring what is referred to as "intelligent compaction," which is the augmentation of machine-integrated compaction measurement systems with mapping systems that tie measurements to their locations on the site, as well as provide data for later analysis and documentation.

THE EVOLVING HISTORY OF COMPACTION MEASUREMENT



historic methods

- Process Control/Method Specifications
- "Point-by-point" testing
  - *Troxler/Nuclear Gauge*
  - *Penetrometers*
  - *Deflectometers*
  - *Plate Load Tests*

Relatively accurate

Typically samples less than 1% of total work area



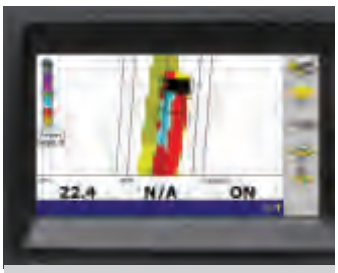
machine-integrated compaction measurement

- Accelerometer
- Machine Drive Power (MDP)

Realtime indications of stiffness or load bearing strength to operator

Difficult to correlate with historic point-by-point measurements

Variability of measurement in certain soil types or applications



intelligent compaction

- Adds satellite mapping capabilities
- Augments machine-integrated compaction measurement by correlating measurements and machine data with the exact location they were taken
- Provides detailed maps to visualize work completed
- Provides data for analysis and documentation

The following highlights some of the most common methods. Keep in mind that each testing method measures soil properties differently (e.g. density, strength, etc.) and as such, attempts to correlate between the tests should be done with a physical understanding of the differences. Further, the very nature of the point-by-point test procedure requires assuming that the results of the point test are valid across untested neighboring areas or the entire jobsite.

Field Measurement Methods (historic) – *Measurements related to Soil Density:* Two basic methods are utilized to measure soil density in the field:

1. **Measure the response of soil to radioactive particles with a device called a Troxler or "nuclear density gauge.**" These devices have become the most common method of measuring soil density in the field, but they have the disadvantage of requiring a license to use and periodic monitoring and testing of the nuclear radiation source in the device. Paperwork and permits are often required to transport the devices.

   The device provides an indication of moisture content at a depth of about 50 mm (2 in) and density at a depth of 300 mm (12 in) or less by emitting neutron and gamma ray radiation, respectively, into the soil being tested. The test is quick and can be performed without disturbing the material. Best results are obtained in homogeneous soils.

   There are three basic procedures of measuring density with a nuclear gauge—direct transmission, backscatter and use of the "air gap method."

   The *direct transmission method* gives the best accuracy, least composition error and least surface roughness error. It can be used for testing over a range of depths from 50-300 mm (2-12"). The most important aspect of the direct transmission method is that the operator has direct control over the depth of measurement.

   The *backscatter method* eliminates the need to create an access hole in the compacted soil because the unit rests on the surface. However, accuracy is compromised and composition errors are likely. This method works best in shallow depths of 50-75 mm (2-3 in).



A Troxler, or nuclear density gauge

   The *air gap method* offers an improvement in composition error and can be used in either the direct or backscatter mode. The testing device is raised above the test surface to lessen the composition error, but accuracy will still not match the direct transmission method.

   Some limitations for nuclear testing equipment are the precautions that must be observed when handling radioactive material, and the fact that false readings are sometimes obtained from organic soils or materials with high salt and/or radioactive content. Ground vibrations from construction equipment can also be a source for errors in the measurements.



The sand cone method has proven accuracy.

2. **Excavate and weigh a volume of soil and measure the volume of the hole from which the soil was excavated.** Weighing a soil sample is direct so long as an accurate scale is available. Measuring the volume of the hole where the soil sample was excavated is not as straightforward, but a couple of procedures have been developed and used with reasonable accuracy. The following describe two examples:

   Sand-Cone Method – The sand-cone method is a multi-step procedure that is more time consuming than the Troxler or nuclear density gauge method, but it has proven accuracy. It is sometimes used in conjunction with the nuclear method to verify the calibration of the nuclear density gauge. This method focuses on measuring the volume of sand required to fill the void from which the soil sample was removed. Sand easily flows into a void and has consistent density, meaning the sand will not have changed density significantly when transferred from a container into the soil void.

   Water Balloon Method – The water balloon method is also called the Washington Densometer Test. The first three steps of the test—excavating a sample, weighing it and drying it—are the same as performed in the sand-cone method. In this manner, moisture content is calculated.

   However, in place of the sand-cone step to measure the volume of the excavated hole, the Washington Densometer is used. The densometer, a fluid-filled device, is placed over the hole. A balloon attached to the base plate is placed in the hole. A valve is opened on the side of the densometer and calibrated fluid is forced into the balloon. As the balloon is filled, it takes the shape of the hole. The densometer is calibrated so the tester can read the volume of fluid and thus, the volume of the hole.

   The density (wet unit weight) is found by dividing the weight of the excavated sample by the volume of the hole—just as with the sand cone method. Dry unit weight can also be calculated by dividing the wet unit weight by one plus the moisture content.

   Limitations to the water balloon method are: the length of time needed to obtain results and the fact that accuracy depends on the ability of the balloon to conform to irregularities in the sides of the hole.



A Dynamic Cone Penetrometer in use.

Measurements related to Soil Strength and Soil Stiffness – Soil density has been the historical measurement for soil compaction, both to set requirements and to qualify results. However, there is a growing trend to use measures of soil compaction more directly related to the engineering requirements, which are most often soil strength. There are several strength measurements used in the field, some of which (but not all of which) will be described in this guide. The strength measurements used in the field fall into three basic categories:

1. **Resistance of the Soil to Penetration –** The most commonly used soil penetration measurement is a *Dynamic Cone Penetrometer (DCP)*. The test is a measurement of the force/energy or shear/frictional strength required to penetrate a small cone into the soil with a fixed-weight hammer that drops a known distance. This device can be used to estimate the California Bearing Ratio and bearing values in newtons per square millimeter (pounds per square inch) to a depth of 1830 mm (72"). Improper use of the drop hammer, or incorrectly counting hammer blows and the change in depth per blow (or series of blows), will cause measurements to be erroneous. Use in rocky soil also produces questionable results.

   The common size cone used has a 20 mm (¾ in) base area with a 60-degree included tip angle. The cone is penetrated by a falling weight striking a plate that then imparts the dynamic force to the cone, causing it to penetrate the soil. Data is recorded as the penetration distance per blow. In harder soils, several blows will be needed to penetrate the cone and the number of blows and distance of penetration is recorded.

   The DCP has the advantage of measuring soil strength to greater depths than other methods. A cone can be penetrated 1 m (39") and beyond, although the friction of the soil on the cone penetrometer shaft can influence readings— particularly beyond a meter of depth. The DCP will not work on very hard materials, such as a lime-stabilized soil after curing. It is physically intensive and can be tiring after repeated use.

2. **Response of Soil to a Falling Weight –** The underlying principle is: stiffer soils will cause more rebound when a weight is dropped onto the surface. One such device is the Light Falling Weight Deflectometer (LWD). A heavier version is the Falling Weight Deflectometer (FWD).

   Light Falling Weight Deflectometer (LWD) measures soil stiffness or elastic modulus of the surface of construction layers by dropping a fixed weight a known distance onto a base plate equipped with accelerometer(s) and other sensors. By detecting the deflection or motion of the base plate, the dynamic modulus of elasticity of the soil is calculated in megapascals (MPa) to a depth of about 150 mm (6 in). Use requires the ground surface to be smooth and flat. Tests show that deflectometers from different manufacturers will produce differing results for the same soil condition.



A light falling weight deflectometer

Falling Weight Deflectometer (FWD) is a larger version of the handheld device discussed previously. The FWD is trailer mounted and not as easy to transport, but it uses the same principles to strike a much larger plate, creating a larger zone of influence and measuring to greater depths. Whereas the LWD can measure to a depth of 150 mm (6"), the larger FWD version is capable of energy that may easily transmit 1 m (39") into the soil, more closely resembling the impact created by a vibratory compactor.

3. **Resistance of Soil to a Static Weight –** This method is similar to the falling weight tests because it measures the stiffness of the soil from the soil surface, but different because it uses a static pressure rather than a dynamic force. Because the purpose of soil compaction is to produce a stiff soil, you need to use a lot of force to effectively deflect the soil with a static pressure. To measure soil stiffness requires a plate size approximately equal to the desired depth of measurement—this means a plate closer to 30 cm (12") in diameter is needed rather than a plate in the centimeter- or inch-size range.

   Plate Load Test – Several plate load test devices are available with various plate diameters. The larger the diameter of the plate, the more force needed to create measurable downward motion of the plate into the soil. As the plate is loaded, the deflection for a given force is measured and plotted. This gives the soil modulus (stiffness) and indicates soil load-bearing capability. Large plate load tests require several thousand pounds of force to drive the plate—typically 300 mm (12 in) diameter—downward into the soil. Available construction equipment often is used as the load source. Plate load test measurements are directly used in pavement thickness design.



Preparing a plate load test



A proof roller

Other Field Measurements – Finally, other field measurement methods that measure compaction, but do not specifically fall into the categories outlined previously, are utilized. These methods have been used for many decades on their own or in conjunction with other tests to further assess the capability of a soil to support a load. Many are still in use today.

1. **Proof Roller –** Proof rolling has been commonly utilized in North America as an alternative method to spot testing in order to directly identify how the compacted surface will respond to an applied load. The test can identify soft spots and ensure a uniform ability to support a load. The process involves rolling a heavy mass on wheels over a compacted surface and measuring the depth of rutting or deflection. Excessive rutting reveals unsuitable compaction.

Proof rolling is not as direct a measure of soil strength as some methods previously described, but it may be a more direct measure of compaction quality when the purpose of compaction is simply to increase ground stiffness for structures such as roads and parking lots. If the soil base does not rut under the weight of a loaded proof roller, then it should not deflect under traffic; that is, assuming that the road is well drained and remains stable over time.

Of all the traditional procedures for testing compaction, proof rolling is the method that provides the greatest testing sample size, as it economically tests a much greater footprint than spot-testing single locations can.

2. **Portable Soil Moisture Oven –** The portable soil moisture oven is a device used to provide a measure of soil moisture content to correlate with the load-bearing measurements taken by the other devices. Measurements from the other devices discussed here are not valid without knowing the soil's moisture content.

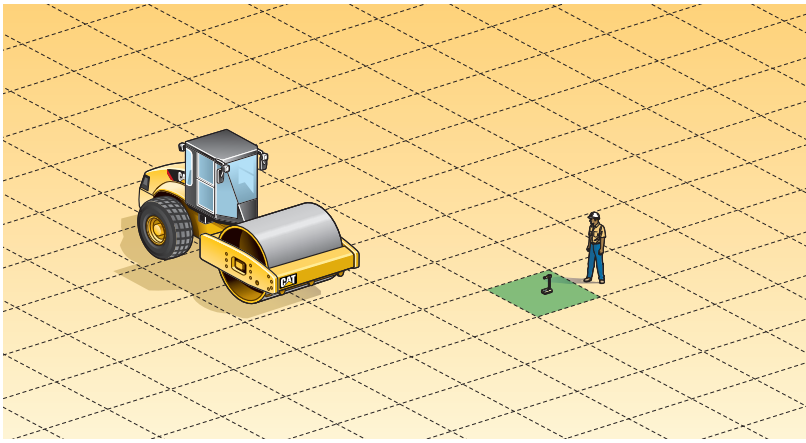


A portable soil moisture oven

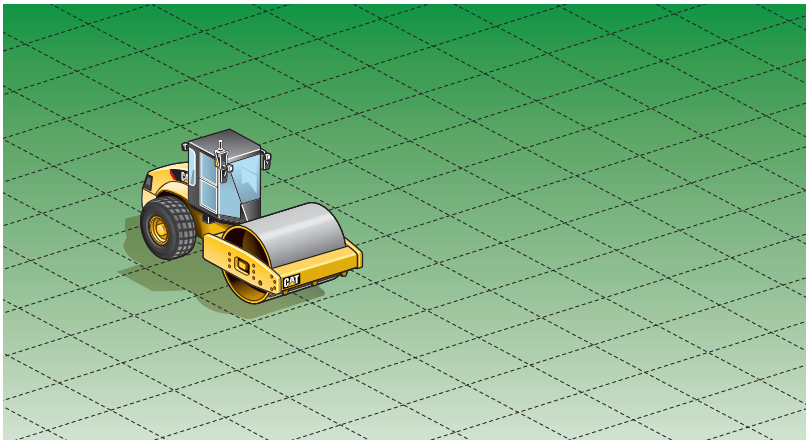
Machine-Integrated Compaction Measurement Methods – Traditionally, compaction has been largely an exercise of trial and error. Operators, lacking dependable data about the state of the soil, rely on their experience and senses to judge when compaction is complete. Or, a method specification provides a procedure determined by engineers to produce satisfactory results if followed properly. The compacted soil is then spot-tested for a predetermined area. Any test results found to be insufficient result in rework for that area. Alternatively to spot testing, some areas utilize proof rollers, typically a heavily loaded trailer or a dump truck. As discussed previously, the proof roller is towed over the site and the depth of the wheel ruts indicates areas where compaction is poor.

Regardless of the method of checking compaction quality, operators are essentially guessing about the quality of their work as the project progresses. And the method of verifying quality utilizing spot checks and proof rollers tests a proportionally small area of a much larger total area that has been compacted. This leaves open the possibility that many areas may not be adequately compacted, a problem that can become quite costly if it causes a future failure of the building or road.

Machine-integrated compaction measurement systems change all of that by empowering operators with data about the state of compaction. Trained operators can use machine-integrated compaction measurement to determine when compaction meets specification, or when an area may have moisture issues. Machine-integrated compaction measurement can alert them to the presence of buried objects that may affect the quality of compaction, such as clay balls, tree trunks or large rocks.



Traditional measurement systems analyze only a small proportion of the compacted area.



Machine-integrated compaction measurement analyzes the entire compaction area.



Machine-integrated compaction measurement systems provide a prediction or indication of soil stiffness.

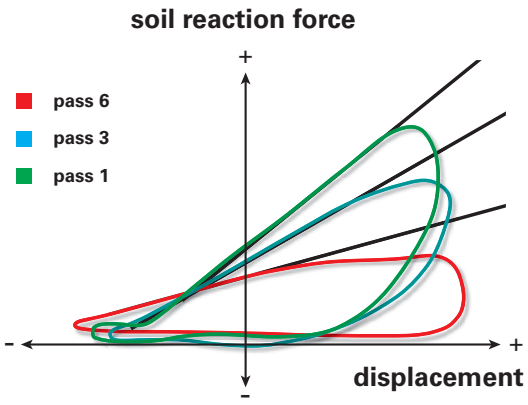
What does machine-integrated compaction technology measure? It is important to understand that these systems do not measure the density of the soil, although the term density is frequently used when discussing results. Because of the variability of compaction work, they do not measure anything directly. What they do is measure a number of factors that provide a prediction or indication about the stiffness of the soil. As previously stated, stiffness is the ability of a certain shape and composition of soil to resist deformation or deflection under the influence of a load. It is a better indication of load-bearing capacity than material density, because certain dense materials can be brittle under load. The material needs to be flexible, but not break.

Two different kinds of measurement technologies are available: accelerometer-based systems and energy-based systems. They make measurements in completely different ways, and correspondingly, what they measure is different as well.

Accelerometer-based Measurement – Accelerometer-based systems are available from most manufacturers. They utilize an accelerometer mounted on the drum to measure the reaction of the soil to being struck by the vibrating drum. Two different methods are employed to make these measurements.

One method is called the Force Displacement method. It uses an accelerometer mounted on the drum axle to measure drum displacement. By measuring the acceleration of the drum and knowing the characteristics of the drum, vibrator and overall machine weight and distribution, a calculation of the force required to achieve a certain displacement is made. The stiffer the soil, the higher the force required to cause the drum to move into the ground a certain depth; or, a constant force will cause the drum to move downward less into the soil as the soil becomes stiffer. Because the contact area of the drum varies with impact into the ground, this method is an estimate and not an exact measurement.

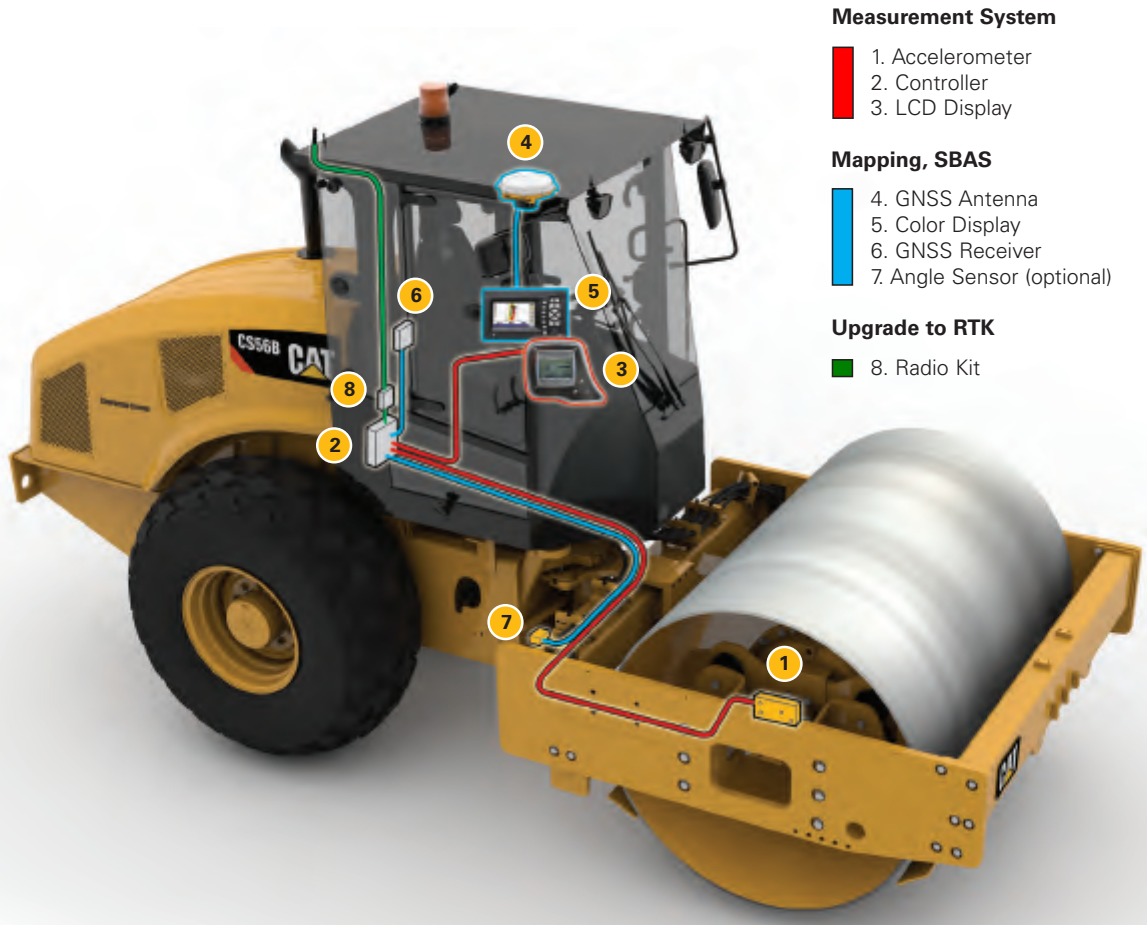
FORCE DISPLACEMENT



The steeper the slope of the curve, the stiffer the soil.



Accelerometers measure the reaction of the soil as it is struck by the vibrating drum.



COMPACTION METER VALUE (CMV)

The second method is called Compaction Meter Value (CMV). Invented by the Swedish company Geodynamik in the 1970s, this method is currently used by Caterpillar as well as several other manufacturers. Rather than attempting to calculate drum displacement, the drum-mounted accelerometer measures the rebound or G-force acceleration at the vibratory frequency and also measures the G-forces at two-times the drum vibratory frequency (this is also referred to as the "first harmonic"). By putting the two values into a formula, a compaction value is calculated that is an indication of soil stiffness, referred to as a Compaction Meter Value (CMV).

Stated another way, the basic principle of measurement focuses on the change of the dynamic response of the compactor as the ground beneath increases in stiffness. Somewhat like a falling weight deflectometer, the accelerometer-based system measures how much the compactor drum is "bouncing" off the ground. Uncompacted soils tend to absorb the vibratory energy, but as the ground stiffens under the compactor with additional passes, energy begins to be reflected off the surface and the drum tends to rebound more rapidly from the vibratory force. Measuring this increased rebound can be translated into an indicator of compaction.

As the soil becomes stiffer, the rebound becomes more pronounced. When the resonant frequency of the ground matches the vibratory frequency of the machine, the soil has reached its maximum stiffness and can no longer accept the compactive energy of the compactor. At this point, the machine will begin de-coupling. Regardless of the measurement method employed, accelerometer-based systems monitor how close the machine is coming to de-coupling. This measurement is called the Resonance Meter Value (RMV) and is used as a gauge of how valid the stiffness measurement is—the closer the machine is to de-coupling, the less valid the measurement.

Accelerometer-based systems measure a deep volume of soil, about 1-1.2 m (36-48 in) deep depending on the composition of the soil and the characteristics of the compactor. This reading is "averaged," so it is not possible to isolate precise stiffness indications at any given depth. But the deep reading is excellent for finding buried objects—in the sub-base, for example—that could affect the quality of the work and the long-term performance of the structure.

One of the drawbacks of accelerometer-based systems is that the drum must be vibrating in order to make measurements. This makes accelerometer-based systems unsuitable for use on cohesive and semi-cohesive soils due to the dampening effect of those materials. For this reason, accelerometer-based systems are ineffective when used on padfoot compactors or in applications when only static compaction is utilized, as the measurements cannot be taken without vibration.

Another drawback of accelerometer-based systems is the depth of measure. As described above, depending on soil type and stiffness of the material being worked, the depth of measurement can be as much as 1.2 m (4 ft) and clearly much deeper/thicker than any one lift being compacted. Thus, you are getting an averaged stiffness value over a number of lifts, or also including the sub-base material stiffness.

CMV - HOW DOES IT WORK?

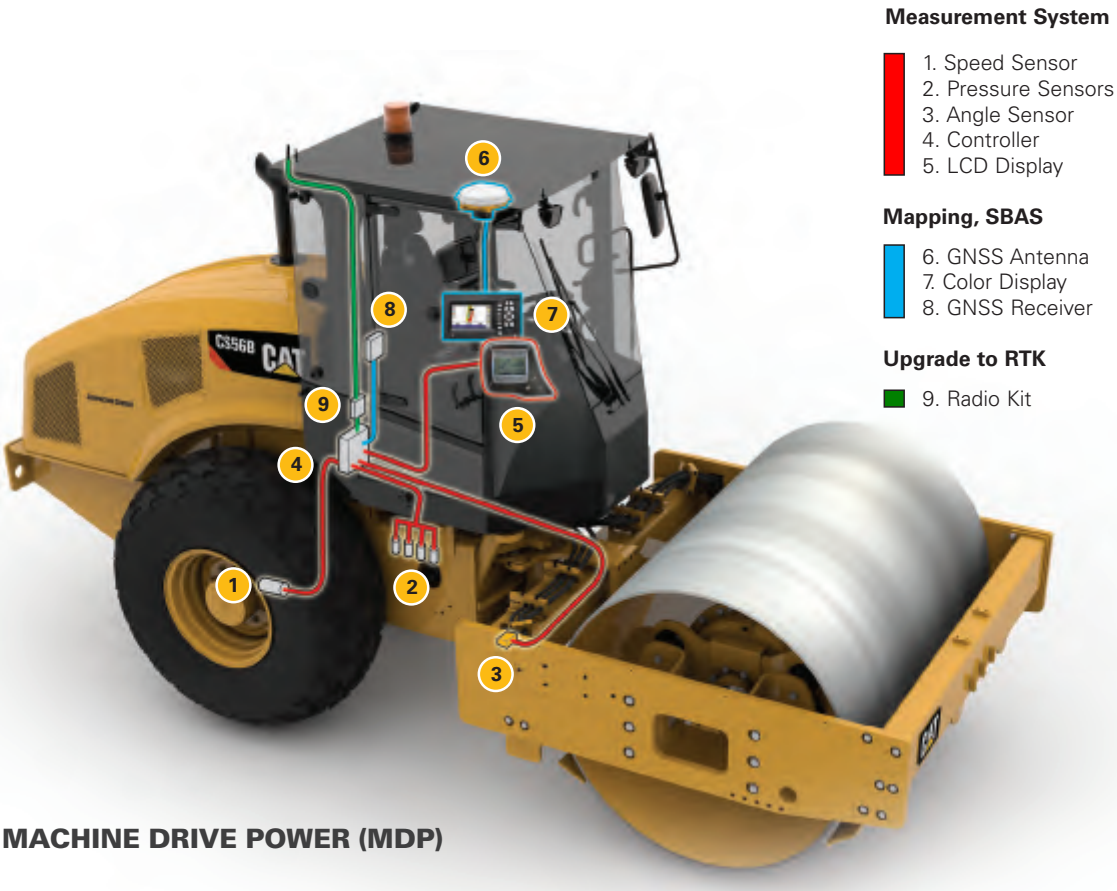


Vibratory energy is imparted on the soil by the vibrating drum.



The material vibrates in response, which is detected and measured by the accelerometer.

CMV calculates a theoretical indication of soil stiffness.



MACHINE DRIVE POWER (MDP)

Energy-based Measurement – The other measurement technology available today measures the amount of rolling resistance the soil compactor encounters as it rolls across the ground. It works on the principle more energy is required to overcome the rolling resistance of loose soils than dense soils. With additional passes, the soil compacts, increasing in stiffness and load bearing strength. As the material offers gradually less resistance, the compactor will require less energy to propel itself over the compacted area. The rolling resistance, and the amount of energy necessary to overcome it, can therefore be correlated with the stiffness of the material. Only Caterpillar currently offers energy-based compaction measurement technology, called Machine Drive Power (MDP).

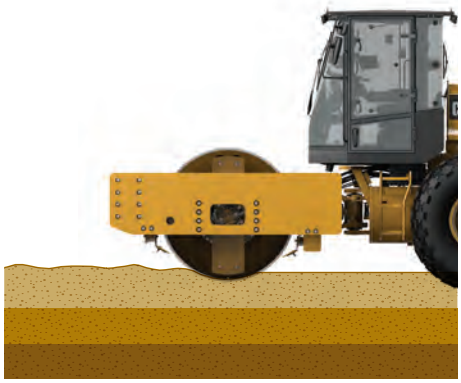
Energy-based measurement has many advantages. Using Machine Drive Power is similar to proof rolling: the less a wheel sinks into the ground, the less energy is required to move the wheel across the ground. As such, a strong correlation exists between MDP and rut depth from proof rolling tests. There is also a high correlation between MDP and soil stiffness, but perhaps the greatest advantage to energy-based measurement is that it is a more tangible and direct calculation and indication of a soil's ability to support a load. If the ground is sufficiently stiff enough to minimize energy transferred from the compacting machine, then it is sufficiently strong to meet the requirements for soil compaction. This relationship, which can be measured for both vibratory and non-vibratory compactors, is the underlying reason to compact soils.

Because the energy-based method of compaction measurement does not require the measurement of vibratory energy in the calculation of soil stiffness, it works well on all soil types including cohesive and non-cohesive soils. It works as well on a padfoot drum compactor as it does on a smooth drum compactor. It functions when the vibratory system is active or inactive. For these reasons, energy-based systems are much more versatile and can be used for more applications than accelerometer-based systems.

Further, energy-based systems do not measure as deep as accelerometer-based systems, about 30-60 cm (12-24 in) deep depending on the soil composition and the characteristics of the compactor. This depth is closer to the depth of a typical lift, so it is measuring the soil that it is compacting rather than an average of a few lifts and/or the sub-base material below the lift being compacted. This depth also is more comparable to the testing depth of portable testing devices, which allows contractors to correlate them with greater confidence.

There are some disadvantages to energy-based systems. They do not measure as deep, so they are sometimes not as adept at identifying buried objects or areas of poor compaction in the sub-base as accelerometer-based systems. Also, when operating an energy-based system with the vibratory system engaged, data warning the operator that the machine is about to de-couple cannot be provided.

MDP - HOW DOES IT WORK?



Propelling over soft ground requires more energy.



Propelling over stiff ground requires less energy.

MDP measures the energy necessary to overcome rolling resistance, a more tangible and direct measurement of soil stiffness.

Factors Influencing Machine-Integrated Compaction Measurement Results – As noted earlier, vibratory compaction is a complex process in which many factors influence the overall compaction force required to achieve compaction/density targets. With machine-integrated compaction measurement technology, operators produce data on 100 percent of the surface compacted, compared with less than 1 percent using traditional portable testing devices/methods. This gives the operator the ability to locate poorly compacted areas or hard spots and take steps to remediate them, which leads to a more homogeneous end result and greater quality.

Accordingly, there are a number of factors that influence machine-integrated compaction measurement results and their correlation to known field test data, regardless of the technology used.

The impact of each factor must be understood in order to use machine-integrated compaction measurement technology to its fullest capabilities. Knowing how each factor influences the measurement will help the contractor perform a consistent measurement with the least amount of variation. To this end, it is beneficial to have a broad understanding of three primary factors:

- Proper preparation of the test site
- Test method and data collection
- Machine parameters

Test Site Preparation

Proper preparation of the test site and tight control of test material conditions are of utmost importance. Failure to ensure homogeneity of the material and uniformity of compaction in both subgrade conditions and raw materials can directly impact the quality and accuracy of the test data.

1. **Base or Sub-base Construction Using Multiple Materials –** Oftentimes, a variety of materials are used in the construction of the base or sub-base. For instance, a hard, crushed-rock base may abut a base constructed of relatively soft clay. When covered with a thick layer of gravel—perhaps 1 m (39 in)—the CMVs measured over the rock base will be substantially higher than the measurements over the clay base. Testing the surface density with a portable testing device would result in nearly identical values, because the devices cannot measure deeper than the surface gravel. Base stiffness has a profound effect on accelerometer-based measurements, but can also have a limited effect on energy-based measurements.

   A stiffer base or sub-base causes the fill material above it to compact easier and to a higher degree. Because of this, Caterpillar recommends that the lowest level of excavation be mapped or proof-rolled to determine its condition is before new fill material is brought in, spread and compacted. It may be necessary to remedy some areas that are substantially harder or softer than the majority of the worksite. Uniformity of compaction is the goal and it requires a uniformly compacted base or sub-base of uniform bearing stiffness.

2. **Soil Type –** The type of soil being compacted has a big influence on the CMV or MDP results obtained.





Preparing a test site

   The reason for this is that the internal soil spring rate and damping rates are very different for fine soils and granular soils. This affects how the soil responds to the vibratory input and, in turn, affects how the measurement is made.

3. **Moisture Content –** Any testing or production compaction should be performed over known, acceptable ground conditions. It is not advised to test over a wet "sponge" or water saturated material. Moisture content of the soil is a variable with enormous impact on compaction results, and will be regularly encountered on the jobsite. It is also a primary cause for variation in CMVs for the same soil type. The contractor does have some control over this factor and can either add water with a water truck and mix it in, or till up the soil and dry it out. For the most efficient compaction, every soil has optimum moisture content. For sandy soils, it ranges from 4 percent to 12 percent and for clay soils it ranges from 9 percent to 22 percent.

Measurements from a machine-integrated compaction measurement system—indications of soil stiffness—are affected by moisture in the soil because of how it fills voids between the soil particles. If there is more air in the voids, the compressibility of the air lowers the reading. If there is more water, its relative incompressibility will result in a higher reading. At some point, so much moisture is present that it acts as a lubricant between particles, allowing them to slide over one another. This will lower the CMV value again.

One benefit of using machine-integrated compaction measurement technology is that the moisture content of a soil can be detected indirectly by how well the soil is packing. For example, a compactor running on granular soils will take more passes to reach a target value as the soil dries out: the drier the soil, the more passes it takes. The operator can track this visually on the display screen, and can call for a water truck to come in and treat specific dry areas. On wet soils, the compactive force of the compactor can cause water to migrate from lower layers to upper layers, resulting in decreased readings. The operator can call for a machine to open the ground up, allowing excess moisture to evaporate.

4. **Hidden Objects Buried in the Soil –** When compacting what appears to be a uniform lift of material, sometimes the machine-integrated compaction measurement in a small area will be substantially higher or lower than the surrounding area. The first response may be to check the area using a surface-reading type of instrument, such as a nuclear density meter or a light falling weight deflectometer. This type of check will rarely produce the same measurement variation as the integrated measurement, because these types of devices do not measure to the same depth.

   A likely cause for a localized high machine-integrated compaction measurement reading would be a large rock or piece of concrete buried in the soil, perhaps a few feet down. A localized low reading could be due to a large clay ball or rubber tires buried in granular aggregate. The variations in integrated compaction measurements are real, but they require excavation of the area to determine just what the object may be. This may not be practical.

   A probe using a long rod driven into the ground or a dynamic cone penetrometer may give some insight into the nature of the anomaly without having to excavate. A decision must be made if the variation will cause problems with the finished road surface during later use. Variation in load bearing strength (stiffness) of the supporting soil can cause stress in the pavement, which leads to a shortened life cycle.





WARNING: When working on sites where there is a possibility of the existence of unexploded munitions buried beneath the surface, it is recommended that a metal detector be used prior to excavation and compaction.

Test Method and Data Collection

1. **Correlation With a Known Field Test Method –** Contractors who are unfamiliar with machine-integrated compaction measuring systems naturally want to relate the data/measurements to a known field compaction test methodology. Troxler (nuclear) density and sand cone density tests have been industry standards for years and show some correlation, but they do not measure the same soil properties or soil depths that the machine-integrated measurement systems do.

   The field compaction test method utilized can directly impact the level of correlation with a machine-integrated compaction measurement value. The effective volume of soil measured by the machine-integrated compaction measurement technology (either CMV or MDP) is substantially greater than that of a typical field measurement method. None of the known field compaction test measurement devices measure to the same depth, volume of soil or even necessarily the same soil properties.

   Due to the portability and relative ease of use, the following are commonly used field test methods:

   - Dynamic cone penetrometer – measures to a greater depth, but is indirectly measuring the shear strength and friction of the soil.
   - Lightweight falling deflectometer – has an effective volume of material tested of about 1 percent of what a machine-integrated system would measure. As such, any non-uniformity of the materials tested would be exacerbated and made even more pronounced by the difference in volume of material tested.

Correlations between the data from measuring devices and machine-integrated measurement systems are variable, depending on which tests and how many tests are performed. When attempting to correlate machine-integrated test data with a known field test, the most satisfactory data correlations result with the use of a large plate load test or falling weight deflectometer.

The final thing to remember when utilizing machine-integrated compaction measurement technology is that what you are measuring—an indication of soil stiffness—is changing while you are measuring it.



As a result, you cannot go back and get the exact same reading you obtained with a previous pass. This can pose a problem for agencies trying to verify compaction measurement values (CMV or MDP) with some other known field test for compaction.

Post-compaction field tests with portable devices can be repeated because they are non-obtrusive and do not disturb the soil. A machine equipped with an integrated compaction measurement system alters the soil on every pass due to its weight. The plate load test is the one test that has a similar effect on the soil as the soil structure itself is changed during the measurement.

Machine-integrated compaction measurement values become more repeatable as the soil nears its final maximum compaction state. At that time, there will be very little change in compaction measurement results between passes. However, if the soil structure is somewhat fragile, the compaction values will bounce up and down because the soil structure builds up to a certain level, then breaks down on the next pass. This is called "de-compaction," and it occurs frequently in some types of granular soil.

2. **Amount of Test Data Recorded –** A narrow range or limited number of measurements can also impact the level of correlation seen with machine-integrated compaction measurement technology. It is recommended that you utilize a well-developed range of field compaction test results to compare to the integrated results and avoid single-point correlations, which do not offer enough information to interpret appropriately.

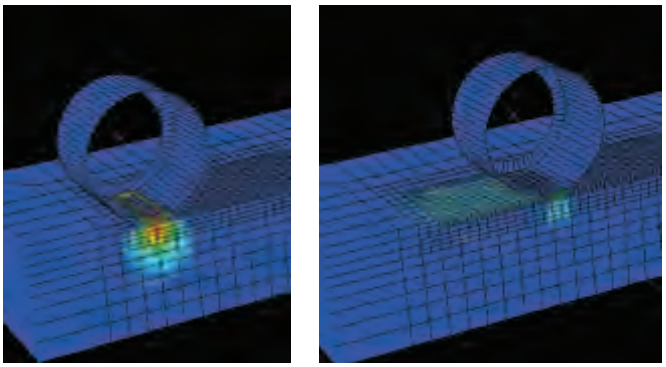
3. **Data Collection –** Uncertainty in spatial pairing of field test results to the machine-integrated compaction measurement values can cause variation and poor correlation. It is recommended to take special care to pair relative data points or to utilize a compactor with GNSS mapping/data collection capabilities in order to associate position detail on the test site to correlate appropriately with field test data.

Machine Parameters

It is well known from a development standpoint—as well as from experience in the field—that machine-integrated compaction measurement technology is sensitive to certain machine operating parameters. Understanding what the system is measuring makes it easier to see how misuse of some of these factors can cause misleading data/output.

1. **Amplitude –** If the drum is at high amplitude, the effect of the drum vibration is transmitted further into the soil. This depth of vibration changes the machine-integrated compaction measurement because the amount of soil being measured is greater. Also, the likelihood increases that the soil is of a different structure and type deeper in the earth.

   If you are looking for information on the variability of soils deep into the ground, high amplitude should be used. If you are interested only in the uppermost layers, lower amplitude is recommended. Even at low amplitude, the soil measurement depth can be a meter or more for CMV results.



High amplitude is transmitted further into the soil.